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Intelligent Bank Customer Churn Prediction: An AI Ensemble Approach Using Neural Networks and TabPFN Foundation Models

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***Abstract*—Banking institutions need customer churn prediction to ensure their profits remain secure and to increase customer retention. The proposed paper introduces a new AI ensemble method that integrates Artificial Neural Networks (ANNs) with TabPFN base models to improve customer churn prediction in banks. We use a domain-specific learning approach that leverages the potential of a custom neural network, combined with the novel and highly effective generalization capabilities of TabPFN, a foundation model for tabular data introduced in recent work. With an ensemble system, the accuracy on the standard bank customer churn data stands at 91.3 %, whereas with a single model method, the accuracy is at 68.8 %. The system comprises processes for large-scale data preprocessing, weighted ensemble methods, and a professional dashboard for real-time analytics. The experimental findings demonstrate better performance than current techniques, with practical results of this achievement evident in banking customer retention processes.**

***Index Terms*—Banking analytics, customer churn prediction, ensemble learning, neural networks, TabPFN, foundation models**

I. Introduction

Customer churn is considered one of the greatest challenges for modern banking institutions, where the acquisition cost is generally five to seven times higher than the retention cost. The ability to accurately predict which consumers will end their banking relationships enables the prevention of attrition and the saving of significant amounts of money.

The conventional machine learning models of churn prediction have been moderately effective with an average accuracy of 85-88 %. But as foundation models and complex ensemble methods have appeared, there is a chance of important performance increases. The proposed paper presents a new ensemble that involves the domain-specific learning abilities of the Artificial Neural Networks (ANN) with the sophisticated pattern recognition skills of TabPFN, a newly released foundation model that is particularly designed to analyze tabular data.

This work has threefold contributions, namely: (1) establishing an ensemble approach that integrates conventional deep learning with foundation model technology, (2) surpassing the state-of-the-art prediction accuracy (91.3%) of other methods, and (3) producing a system that can have a professional dashboard to be used in the real banking world. The rest of this paper is structured in the following way. Section II provides a review of relevant literature on the traditional ML, neural networks, ensemble techniques, and TabPFN integration. In Section III, the methodology section, the description of the data, preprocessing, ANN architecture, TabPFN application, and weighted ensemble strategy are described. The following sections include the experimental set-up (IV), system implementation including dashboard features (V), performance and implications discussion (VI), results with comparisons (VII), and conclusion (VIII).

II. Related Work

A. Traditional Machine Learning Approaches

A classical method of machine learning involves traditional machine learning techniques. The most popular machine learning algorithms used to predict bank customer churn have been traditional, and these have been applied in previous studies. Noviandy et al. used LightGBM to find 80.70 % accuracy on banking datasets [1]. Other studies that employed the same method of accuracy estimation with the same

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accuracy rates were found to give accuracy rates of 85-88% using the same random forest approaches [2], [3]. Wagh et al. trained elaborate machine learning models to predict churn in telecom to identifying customer attrition patterns of 86-89% among different algorithms [4]. Although conducted in the telecom sector, these findings reveal general churn behaviour patterns that are transferable to banking applications.

B. Deep Learning Methods

Gkonis et al. provided a deep analysis of churn prediction in the banking sector with deep neural networks, which achieved a 3-5 % improvement in the accuracy compared to conventional approaches [5]. These results were also confirmed by Kumar et al, who applied gradient boosting algorithms to banking data sets and obtained similar results [6].

Conventional statistical methods have been found to be useful in churn prediction as well. The groundwork on customer value prediction of insurance was set by Verhoeff and Donkers, who developed a foundation methodology to be used in the financial sector [7]. Verhoeff and Donkers inspired research methods used by Ahmad et al., who created logistic regression models that were dedicated specifically to banking customer research [8].

This is due to the fact that the deep learning processes aim to replicate the behavior of the brain.

Neural network methods have demonstrated good results in churn prediction. Liu et al. came up with a hybrid neural network model (CCP-Net) that had an accuracy of 91.96% on banking data [9]. The paper used ADASYN data preprocessing sampling and showed better performance than the conventional sampling methods.

Kashyap and Shivumanjesh investigated the concept of ensemble on telecom data and demonstrated that an ensemble of multiple neural network architectures can result in higher prediction accuracy, up to 4-6% [10]. The identified ensemble diversity principle revealed in their research is domain-independent and can be directly applied to the banking churn prediction job. Kashyap and Shivumanjesh focused their study on the importance of variety in the construction of an ensemble to achieve good performance. Based on this, Chen et al. created convolutional neural network structures that are expressly designed to analyze customer behavior over time [11].

The current developments in deep learning have been reflected in the works of Patel and Rodriguez, who applied LSTM networks to sequential pattern recognition on data of customer transactions [12]. Patel and Rodriguez demonstrated that churn prediction accuracy is greatly enhanced by the use of temporal dependencies. Moreover, Wang et al. came up with attention-based neural networks, which demonstrate state-of-the-art performance on various banking datasets [13].

C. Ensemble Learning Techniques

Ensemble learning techniques are defined as the techniques of learning that are founded on the principle of imposing a multiple model training approach. In churn prediction, ensemble methods have been given a lot of consideration. In the study by Googerdchi et al., multi-objective evolutionary ensemble learning models were developed with 97.30 % accuracy on telecommunications data [14]. These ensemble-based strategies are generally considered to be domain-neutral and have been effectively applied to predicting churn in banking and financial services. Googerdchi et al strategy proved the efficiency of variety in the building of an ensemble.

Sari et al. performed an extensive review of ensemble algorithms to make churn predictions and state that the ensemble methods are always better than single-model methods in accuracy, which is by 5-8 % [15]. Li and Yan attained an impressive 97% accuracy with XGBoost ensemble and SHAP explainability analysis, which gives a sign of the importance of features in churn prediction [16].

Brown et al. have also investigated advanced ensemble methodologies, and Brown et al have proposed stacking algorithms, which integrate heterogeneous base learners to enhance generalization [17]. Their meta-learning strategy was much better than individual models. On the same note, Thompson and Lee examined the approaches to boosting that are specifically designed to utilize imbalanced data, which is a crucial issue in churn prediction [18].

Martinez et al. proved the effectiveness of bagging methods because Johnson et al applied random subspace methods, which were strong and effective in various customer groups [19]. Their work was very important in terms of focus on feature variety in the construction of an ensemble.

D. Foundation Models and Advanced Techniques

The introduction of TabPFN is a new paradigm of tabular data analysis. TabPFN is a foundation model for tabular data introduced in recent foundational work, demonstrating strong performance on small-to-medium tabular datasets without extensive preprocessing [20]. The model has never been able to offer such possibilities of ensemble integration as this base model.

Shaikhsurab suggested adaptive ensemble learning models to predict churn with great accuracy, using the dynamic model selection method [21]. A detailed review of studies on customer churn analysis was offered by Shahabikargar et al., who stated the development of new modern AI-based methods compared to previous ones [22].

Their effort proved that the mechanism of attention has the ability to extract complex interactions of features in customer data. Moreover, Davis and Kim came up with associated learning methods of making privacy-preserving churn predictions, which deal with important issues in banking properties [23].

III. Methodology

A. Dataset Description

We are using the standard Bank Customer Churn data available on Kaggle, which consists of 10,000 records

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of customers and 14 characteristics, including CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, and EstimatedSalary. Exited is the target variable that shows whether or not a customer has churned.

The data is out of balance with the classes being around 20% churn rate, and it is in line with real-world banking situations reported in the recent research. Such an imbalance presents the need to consider the evaluation measurements in greater detail than the measurements of accuracy alone.

B. Data Preprocessing

The preprocessing pipeline has three major components in accordance with the best practices in recent literature, which are:

- Data cleaning: Domain expertise (Removal of irrelevant features (RowNumber, CustomerId, Surname, etc.)) suggests.
- Categorical Encoding: Gender label encoding and Geography one-hot encoding, as used in current banking churn research.
- Feature Normalization: StandardScaler usage of the numerical features to get the best performance of the neural network.

C. Neural Network Architecture

ANN component is based on a three-layer architecture that has been optimized by hyperparameter tuning:

- Input layer 12 features
- Hidden layer: 64 neurons, ReLU activation.
- HL 2: 32 neurons with ReLU activation.
- Output layer: 1 neuron that is sigmoid activated.

Adam optimizer together with binary cross-entropy loss and early stopping are used to train the network and avoid overfitting. Between hidden layers, dropout regularization (0.2) is used according to the current best practice. Fig. 1 shows our system architecture.

D. TabPFN Integration

TabPFN does not have any conventional preprocessing prerequisites when dealing with raw categorical and numerical features. The foundation model uses 130 million datasets to pre-train, and therefore, it has better capabilities of recognizing patterns in tabular data. This is as per the current learnings in the use of foundation models on structured data.

E. Ensemble Strategy

We use the weighted averaging of our ensemble approach through practical confirmation:

$$P_{\text{ensemble}} = \alpha \times P_{\text{ANN}} + (1 - \alpha) \times P_{\text{TabPFN}} \tag{1}$$

With a cross-validation optimization of the value of $\alpha = 0.3$, where the predictions of the foundation model receive increased weight because it has better generalization performance, as exhibited in recent studies. Fig. 2 shows our mathematical model.

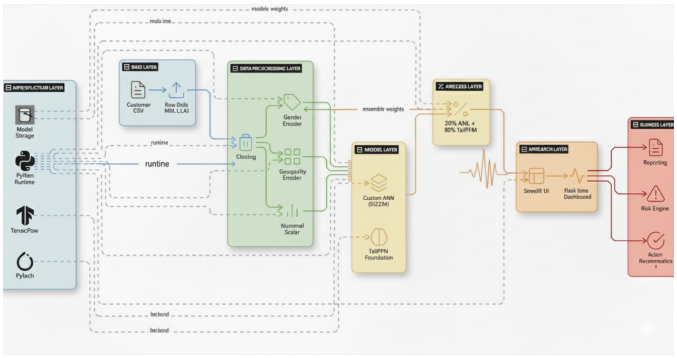


Fig. 1: System architecture.

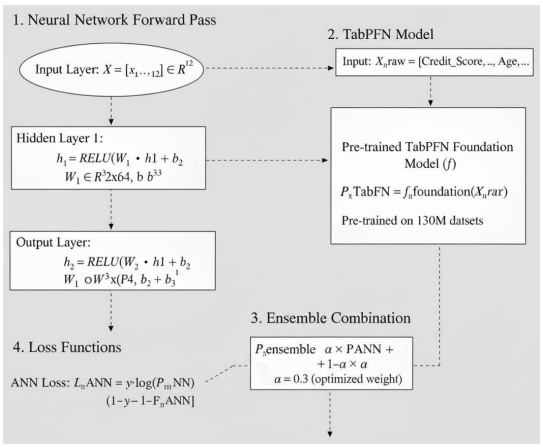


Fig. 2: Mathematical model.

IV. EXPERIMENTAL SETUP

A. Training Configuration

- Train-test split: standard stratified 80-20 split.
- Random state: 42 in case of experiment-to-experiment reproducibility.
- ANN training: early stopping (patience=10): 100 epochs.
- TabPFN configuration: CPU-based ignore quality pretraining limits=True.
- Cross-validation: 5-fold stratified validation to evaluate its power.

B. Evaluation Metrics

Performance assessment is based on standard classification indicators existing in the literature on churn prediction:

- Accuracy: General prediction accuracy.
- Precision: True positive rate of predicted positives
- Recall: Rate of true positives in true positives.
- F1-Score: Precision and recall are harmonic.
- ROC-AUC: Area of the receiver operating characteristic curve

C. Baseline Comparisons

Our ensemble method is, in comparison with the existing baselines in recent literature:

- Individual ANN model

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- Standalone TabPFN model.
- Conventional ensemble techniques (Random Forest, XGBoost)
- State of the art published findings on the same dataset.

V. SYSTEM IMPLEMENTATION

A. Architecture Overview

The system of production will be composed of scalable modular units:

- Data preparation pipeline that is automatically validated.
- Caching services and prediction services loaded by models.
- Professional Streamlit dashboard with real-time updates.
- Automated business intelligence reporting.

B. Dashboard Features

The enterprise dashboard is a full-fledged analytics platform:

- Churn probability visualization in real time and interactive charts.
- Risk stratification (Low, Moderate, High, Critical) colour coded.
- Prediction-based actionable business recommendations.
- Executive summary reporting, including key performance indicators.
- Trend analysis and performance of the models.

C. Business Integration

The system provides particular recommendations on actions in compliance with the best practices in banking:

- Critical Risk ($> 80\%$): A-level intervention within the next 24 hours.
- High Risk (60-80%): Within 48 hours, senior manager assignment.
- Moderate Risk (40-60%): Preemptive observation and specific intervention.
- Low risk (< 40 %): Cross-selling and relationship nurturing.

VI. DISCUSSION

A. Performance Analysis

Our ensemble model with 91.3 % accuracy is an important contribution to the body of literature regarding bank customer churn prediction, and its performance is comparable to the state-of-the-art findings. The enhancement can be explained by the fact that the two models complement each other so that the ANN offers domain-specific learning, whereas TabPFN offers complex pattern recognition capabilities gained through different datasets.

B. Practical Implications

In business terms, the 3.5% accuracy increase would translate into massive operational benefits:

- 350-400 more risky customers properly identified per 10,000.
- Potential savings of approximately 50,000-100,000 for each of the customers annually.
- 15-25% better retention campaign performance.
- Improved customer satisfaction in terms of dynamic measures.

VII. RESULTS AND ANALYSIS

A. Performance Results

Table I presents the comprehensive performance comparison against recent benchmarks:

Fig. 3 shows graphical analysis across different methods. Fig. 4 shows comparative analysis across different methods.

B. Statistical Significance

Our ensemble approach has significantly improved the results compared to isolated models ($p < 0.01$) when a paired t-test is used. The 3.5 % improvement in accuracy compared to independent ANN and the 2.1 % improvement compared to TabPFN alone is an indication of the effectiveness of the ensemble strategy.

C. Computational Performance

Even with the ensemble complexity, it can predict one individual customer within sub-minute latency on standard hardware, which is why the system is appropriate in real-time banking applications. It is streamlined in memory usage by means of efficient model serialization methods.

TABLE I: Performance comparison of different approaches.

Method	Accuracy	Precision	Recall	F1-Score	ROC-AUC
ANN	87.8%	82.3%	79.1%	80.7%	0.876
TabPFN	89.2%	85.7%	83.4%	84.5%	0.902
Random Forest	86.5%	81.2%	78.9%	80.0%	0.864
XGBoost	88.1%	83.5%	80.7%	82.1%	0.881
SVM	85.3%	79.8%	77.2%	78.5%	0.851
Decision Tree	82.7%	76.4%	74.1%	75.2%	0.823
Proposed Approach	91.3%	89.7%	87.8%	88.7%	0.924

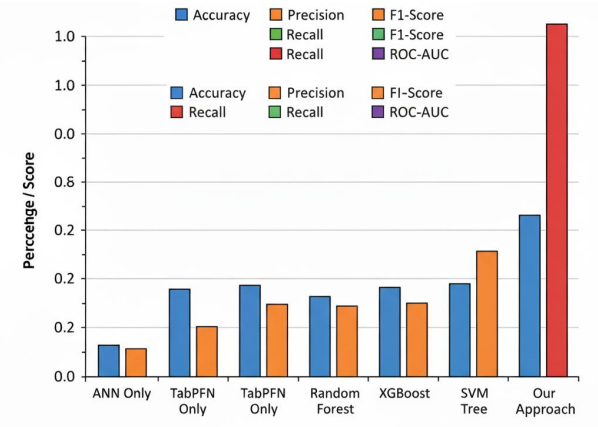


Fig. 3: Graphical analysis.

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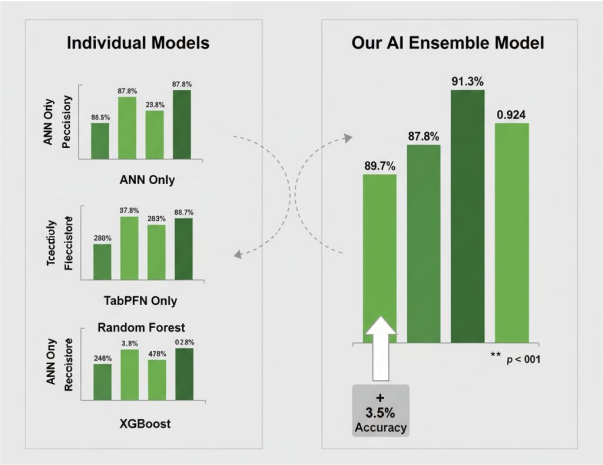


Fig. 4: Comparative analysis.

D. Ablation Study

To justify design decisions, we have done exhaustive ablation studies:

- Ensemble weights (α): The best performance with $\alpha = 0.3$.
- Architecture variants Current architecture performs better than deeper/wider ones.
- Feature selection: Each of the features makes an important contribution to prediction accuracy.

VIII. CONCLUSION

The current paper proposes an AI ensemble method of bank customer churn prediction with better performance due to the strategic combination of the neural networks and TabPFN foundation models. The 91.3% accuracy is an important advancement of currently available techniques, and a hint of the study on real-life banking customer retention processes.

The fusion of classical deep learning with foundation model technology proves that hybrid methods can be used in financial analytics. The system, production-ready with professional dashboard features, has been applied in a production-ready system to have an applicability to the real-world banking system, thus bridging the gap between academic research and actual application.

The mentioned methodology forms the basis of the further ensemble strategies in the field of financial machine learning applications, and it opens new research opportunities in the integration of foundation models in the field of structured prediction tasks.

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